HARF

Literature Review & Project Evaluation

Interactive Speech & Language Support Platform for Children

PROJECT FOCUS HARF is designed to make home practice more engaging and organized through child-friendly activities, specialist-assigned plans, parent progress views, and AI-assisted suggestions reviewed by the specialist. It supports professional care; it does not replace diagnosis or therapy.

1. Literature Review | Evidence relevant to the project

Research points to three relevant directions: Arabic-language digital support, play-based learning with caregiver involvement, and the need to evaluate technology with real users. The following sources inform HARF's design; they do not prove that HARF itself is effective.

SOURCE / TOPIC	KEY FINDING	IMPLICATION FOR HARF
Alzahrani (2024) **Arabic educational apps**	Reviewed 9 studies covering 7 Arabic-language apps for autistic children. The review highlights a small but relevant Arabic-focused research landscape.	Use Arabic-first content and familiar examples; involve speech-language specialists in reviewing activity wording and suitability.
Toma et al. (2024) **Digital/XR interventions**	A systematic review included 28 studies. Nineteen apps were tested with children, while 9 were tested only by developers, underscoring the need for stronger user evaluation.	Plan usability testing and gather feedback from parents and specialists instead of relying only on technical checks.
Deniz et al. (2024) **Parent-mediated play**	A systematic review/meta-analysis included 26 studies, with 21 in the synthesis. Play-based, parent-mediated interventions showed positive pooled effects on social communication and language in preschool autistic children.	Connect home activities with parent support and specialist guidance. This evidence concerns play-based interventions generally, not HARF or digital apps specifically.

2. Research Gap & Project Alignment

The project proposal identifies **inconsistent home practice, repetitive exercises, weak parent-specialist coordination, and limited progress visibility** as the problems HARF aims to address. Its two modes connect to these needs: a free game library with points and achievements, and a specialist-guided mode with personalized plans, assigned activities, results, and parent progress views. AI features are assistive only: suggestions and draft activities require specialist review.

Literature-based takeaway: HARF's strongest rationale is not simply adding games or AI, but organizing repeatable home activities around specialist goals and making progress easier to follow. Clinical effectiveness must be evaluated separately.

EVALUATION, IMPROVEMENTS & ASSESSMENT

3. Feedback & Evaluation | Proposed review plan

AREA	HOW TO EVALUATE
Usability & accessibility	Can children and caregivers understand the screens and complete activities? Is the interface responsive and easy to navigate?
Learning activity quality	Are instructions, audio, Arabic language, and difficulty levels clear and appropriate? Activity content should be checked by qualified specialists.
Role-based workflow	Can specialists assign plans and review results, while parents see only their child's information and relevant activities?
Reliability, privacy & security	Test authentication, role permissions, data privacy, input validation, and core flows across supported devices.

Suggested method: combine functional tests with task-based usability sessions and feedback from the instructor and, where feasible, parents and qualified speech-language specialists. Any testing involving children should use appropriate consent, privacy protections, and specialist oversight. Do not claim clinical improvement from a prototype usability test.

4. Suggested Improvements | Prioritized next steps

- **1. Define activity categories and levels:** organize exercises by target skill (e.g., vocabulary, comprehension, sentence building) and by difficulty, with specialist approval.
- **2. Strengthen home guidance:** give parents short, clear instructions on how to support assigned practice without replacing professional advice.
- **3. Improve accessibility:** use simple language, clear audio/visual feedback, predictable navigation, responsive layouts, and options that avoid overstimulation.
- **4. Protect child data and govern AI:** apply role-based access, collect only necessary information, and label AI output as a draft for specialist review—not diagnosis or independent treatment guidance.
- **5. Validate with users:** run a small, supervised pilot to find usability issues before evaluating educational or clinical outcomes.

5. Final Grading Criteria | Suggested rubric - confirmation with instructor

Documentation & literature review	**20%**
Implementation & required functionality	**35%**
Testing, usability, accessibility & security	**25%**
Presentation, demonstration & teamwork	**20%**
TOTAL	**100%**

These weights are a proposed team rubric, not official DEPI grading weights. Replace them if the instructor provides a formal breakdown.